

DFT’s missing energy: same density, different electron-pair probabilities.

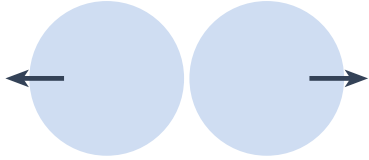
A **functional** maps the whole density to an energy. Kohn-Sham uses **noninteracting electrons at the same density**.



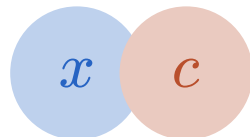
already quantum



V_{ext}
nuclear attraction



E_H
smooth-charge repulsion



E_{xc}
the missing correction

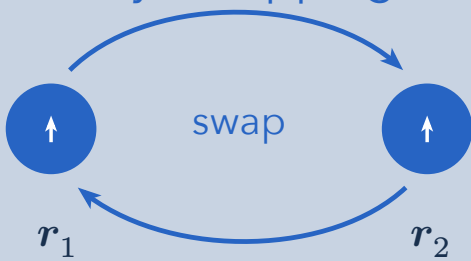
$$E[n] = T_s + V_{\text{ext}} + E_H + E_{\text{xc}}$$

KS kinetic energy

$$E_{\text{xc}} = \underbrace{T - T_s}_{\text{kinetic correction}} + \underbrace{V_{ee} - E_H}_{\text{interaction correction}} = E_x + E_c$$

EXCHANGE

Antisymmetry: swapping fermions changes the sign.



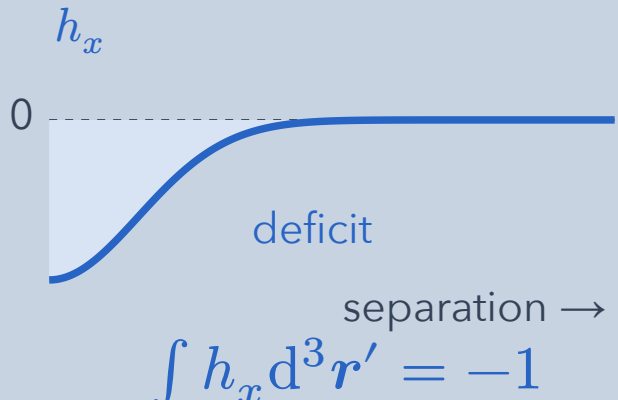
swap

$\Psi \rightarrow -\Psi$
same position $\rightarrow \Psi = 0$

Pauli exclusion • same spin only • one determinant
Opposite-spin pairs have no exchange hole.



same-spin probability

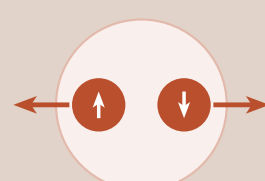


h_x
deficit
separation $\rightarrow$
 $\int h_x d^3r' = -1$

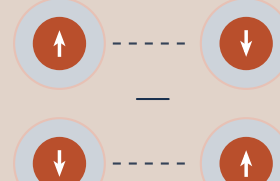
One missing electron. No extra force.
The hole persists without Coulomb repulsion.

CORRELATION

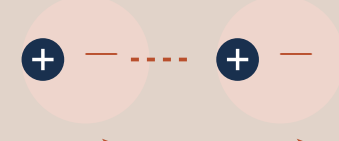
Joint probabilities beyond the KS determinant.



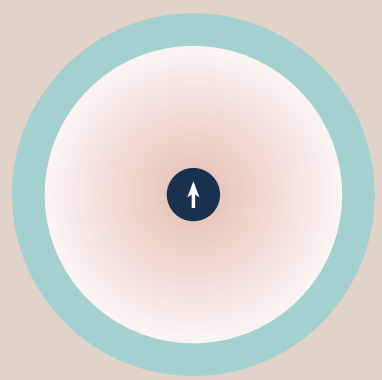
DYNAMIC
short-range avoidance and screening



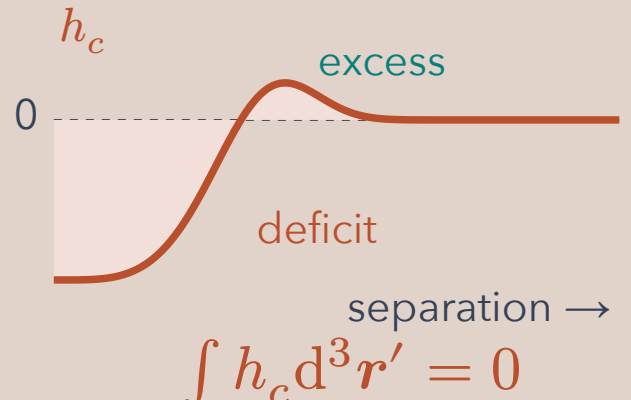
STATIC
stretched-bond singlet: several configurations



DISPERSION
coupled fluctuations, even at zero temperature



change in probability



h_c
excess
deficit
separation $\rightarrow$
 $\int h_c d^3r' = 0$

Same-spin and opposite-spin pairs.
A near deficit is balanced by an excess farther away.

$$E_x = \langle \Phi_s | \hat{V}_{ee} | \Phi_s \rangle - E_H \leq 0$$

Φ_s : antisymmetric KS orbital state. T, V_{ee} : exact interacting values at the same density.

$$E_c = \underbrace{T - T_s}_{T_c \geq 0} + \underbrace{V_{ee} - E_H - E_x}_{U_c} \leq 0$$

Correlation balances reduced repulsion against a kinetic cost.

ONE REFERENCE ELECTRON

Only $N - 1$ others remain.

$$n_{\text{cond}} = n + h_x + h_c$$

net hole: $-1 + 0 = -1$
probability deficit, not an empty cavity

CLASSICAL CHARGES



correlated positions

Correlation: **yes**
Pauli exchange: **no**

ONE ELECTRON
NO SELF-REPLICATION


$$E_H + E_x = 0$$
$$E_c = 0$$

Approximate XC can violate this cancellation.

XC FEEDS BACK INTO THE DENSITY

$$v_{\text{xc}} = \frac{\delta E_{\text{xc}}}{\delta n} \rightarrow \text{orbitals} \rightarrow n$$

Why approximate? Exact XC is formally defined, but no practical general expression is known.

LDA: local electron gas · GGA: density gradients · hybrids: orbital exchange. Ordinary LDA/GGA miss long-range dispersion.

Schematic maps / curves; map center = reference electron. Hole sums are 3D integrals. Fixed nuclei; spin labels suppressed; atomic units; nuclear repulsion added separately.